

TAHSIN NAYEEM SHRESTHA

tahsin.shrestha@stonybrook.edu | linkedin.com/in/tahsin-nayem-shrestha | tshrestha-1114.github.io/tahsin-portfolio

EDUCATION

Stony Brook University, New York, NY, USA.

05/2026

Bachelor of Science, Information Systems Engineering

Minor: Studio Arts

Specialization: Financial System Management

Honors and Awards: Women in Science & Engineering Program (WISE), Dean's List 2025, Dean's List 2023

Cornell University, New York, NY, USA.

05/2023 - 05/2024

Certificate in Machine Learning Foundations and Artificial Intelligence

EXPERIENCE

SBU IT Department, Software Developer, New York, NY, USA.

09/2022 - Present

- Engineered an LLM-powered (GPT-4, Claude) ticket analysis pipeline processing 50,000+ TDX IT service records across multiple academic years, automating classification and summarization workflows that significantly reduced manual categorization effort and improved reporting turnaround.
- Built an agentic multi-step reasoning workflow using GPT-4 and Claude to dynamically triage unstructured ticket data, detect anomalies, and surface macOS/Windows imaging trends enabling faster, evidence-based decisions for campus IT operations.
- Partnered with a cross-functional data analytics team to design standardized reporting methodologies and PostgreSQL data quality pipelines, translating raw IT service data into actionable insights that directly supported data-driven decision-making at scale.

KPMG International Limited, AI Engineer Apprentice, New York, NY, USA.

07/2023 - 12/2023

- Spearheaded a Python-based analysis of 1M+ U.S. Census 2020 records stored in AWS S3, building and deploying regression, ensemble, and neural network models that quantified income disparities in New York State directly informing equity policy recommendations.
- Engineered predictive modeling pipelines with NumPy, Pandas, and Seaborn, using DynamoDB for scalable record management to detect gender-based salary gaps of up to 15% in low-wage sectors, improving model interpretability and decision-support accuracy by 20%.
- Translated model outputs indexed via Amazon Elasticsearch into actionable equity recommendations that drove a measurable 10% reduction in regional pay inequality, validated through continuous model retraining and iterative performance benchmarking.

Division of Information and Technology, Technology Assistant, New York, NY, USA.

09/2022 - Present

- Delivered technical support to 500+ students, faculty, and staff via walk-ins, remote sessions, and on-site troubleshooting across 10+ campus buildings, ensuring 99% service satisfaction and minimal down-time.
- Performed 200+ hardware/software maintenance tasks, including OS imaging, secure data wiping, driver configuration, and device deployment; contributed to maintaining campus-wide print systems with 100% uptime.
- Trained and mentored 50+ new technicians, improving onboarding efficiency by 30% while streamlining documentation and ticket-resolution workflows within the IT service management platform.

PROJECTS

Pathfix App

09/2024

- Mined and analyzed data from over 200 courses and 1000 job skill sets with 95% accuracy, demonstrating data collection and analysis expertise using sources such as Google Jobs and Kaggle. Implemented Python and MongoDB to build the website.
- Optimized data integration with a 30% improvement by incorporating feature augmentation, showcasing skills in data integration, process optimization, and problem-solving.

Edge-Based Developer Feedback Analytics Platform (Cloudflare Workers)

01/2026

- Built a serverless edge analytics application using Cloudflare Workers to aggregate and process developer collaboration data from distributed sources.
- Developed REST API integrations and data pipelines in JavaScript, designing a scalable architecture for low-latency data processing without traditional servers.

Network Traffic & Protocol Analysis Toolkit

12/2025

- Built tools to analyze PCAP network traffic, extracting flow-level statistics across TCP, DNS, and routing protocols. Implemented logic to compute latency, throughput, packet loss, and congestion patterns, diagnosing protocol inefficiencies under varying network conditions.
- Applied systems-level debugging to interpret real-world network behavior and validate protocol performance assumptions.

SKILLS

- Programming Languages:** Python, R, JavaScript, React JS, Java, Ruby, C, Bash.
- Software:** Linux, Scikit Learn, TensorFlow, Pandas, Numpy, PostgreSQL, SQL Lite, NoSQL, AWS (API Gateway, S3, OpenSearch, Lambda), Azure, Django, Node.js, Cloudflare wrangler, Git, REST APIs, CI/CD pipelines.
- Project Management:** Agile Methodologies, Jira, Excel, Microservices Architecture.